

Loading Data-JSON



Learn2
CloudData

Powered by

www.learn2clouddata.com

Sujith Nair

Cloud Data Architect
Snowflake Snowpro Certified

Section Introduction–Loading semi-structured Data

- What is semi-structured data ?
- What is JSON ?
- What are the semi-structured data types supported in Snowflake ?
- LAB–Loading semi-structured Data to Snowflake
 - Loading and selecting data from OBJECT data type.
 - Loading and selecting data from ARRAY data type.
 - Loading and selecting data from VARIANT data type.
 - Flattening JSON data with LATERAL FLATTEN–Single Record
 - Combining JSON data with UNION ALL
 - Flattening JSON data with LATERAL FLATTEN–Multiple Record
- Summary.

What is Semi-Structured data in Snowflake ?

Semi-Structured data is a form of data that does not conform to a rigid, predefined data model like a table with fixed structure of columns and rows but rather contains properties such as Key value pairs(JSON) or Tags (XML) to store or represent the data.

Features of semi structured data:

- Semi structured data formats supported in Snowflake are JSON, Parquet, Avro, ORC and XML.
- JSON & parquet are the most important semi structured data.
- ORC(Optimized Row Columnar) is very similar to parquet and is columnar storage type.
- Data in data lake is generally stored either in parquet or ORC format due to columnar storage.
- Avro is a data format that is used with messaging systems like Kafka and is in binary format. Avro is basically JSON in binary format.
- XML is a file format for storing, transmitting data between applications.
- Applications which are on completely different technologies can exchange data between them with the help of XML.

What is Semi-Structured data in Snowflake ?

```
<?xml version="1.0" encoding="UTF-8"?>
<electricalEquipmentStore>
  <equipment>
    <id>1</id>
    <name>LED Light Bulb</name>
    <brand>Philips</brand>
    <price currency="USD">5.99</price>
    <wattage>9W</wattage>
    <voltage>110V</voltage>
    <category>Lighting</category>
    <lifespan>25000 hours</lifespan>
  </equipment>
  <equipment>
    <id>2</id>
    <name>Electric Kettle</name>
    <brand>Hamilton Beach</brand>
    <price currency="USD">24.99</price>
    <wattage>1500W</wattage>
    <voltage>120V</voltage>
    <category>Appliances</category>
    <color>Stainless Steel</color>
    <warranty>1 year</warranty>
  </equipment>
</electricalEquipmentStore>
```

```
{
  "electricalEquipmentStore": {
    "equipment": [
      {
        "id": 1,
        "name": "LED Light Bulb",
        "price": {
          "amount": 5.99,
          "currency": "USD"
        },
        "wattage": "9W",
        "voltage": "110V",
        "category": "Lighting",
        "lifespan": "25000 hours"
      },
      {
        "id": 2,
        "name": "Electric Kettle",
        "price": {
          "amount": 24.99,
          "currency": "USD"
        },
        "wattage": "1500W",
        "voltage": "120V",
        "category": "Appliances",
        "color": "Stainless Steel",
        "warranty": "1 year"
      }
    ]
  }
}
```

What is JSON ?

JSON is a data exchange format that uses human readable text to store and transmit data in key value pairs.

Features of JSON data:

- Uses key value pairs to store and transmit data.
- Is lot more flexible than structured data and can store hierarchical, nested and schema evolving structures.
- Is a modern and latest format for storing and exchanging data.

What are the semi structured data types supported in Snowflake ?

There are 3 semi-structured data types supported in Snowflake.

- OBJECT
- ARRAY
- VARIANT

What are the semi structured data types supported in Snowflake ?

OBJECT An Object consists of one or more key value pairs where each key is a unique string and value can be provided for this unique string or Key.

```
{  
  "name": "John",  
  "age": 30,  
  "skill": "SQL"  
}
```

```
CREATE TABLE EMPLOYEE_DATA  
(  
  EMPLOYEE_INFO OBJECT  
)
```

```
SELECT  
  e.name,  
  e.age,  
  e.skill  
FROM EMPLOYEE_DATA e
```

```
{  
  "name": "John",  
  "age": 30,  
  "skill": "SQL",  
  "address": {  
    "street": "123 Main St",  
    "city": "Techville",  
    "state": "CA",  
    "zip_code": "90001"  
  }  
}
```

```
SELECT  
  e.name,  
  e.age,  
  e.skill,  
  e.address.state  
FROM EMPLOYEE_DATA e
```

What are the semi structured data types supported in Snowflake ?

ARRAY An ARRAY is an ordered collection of elements and stores a list of values. The elements in the array have a specific sequence and you can access each element by its position (index) in the array.

skills
['SQL', 'Python', 'Java']

```
CREATE TABLE EMPLOYEE_DATA  
(  
  EMPLOYEE_INFO ARRAY  
);
```

```
SELECT  
  skills[0],  
  skills[1],  
  skills[2]  
FROM EMPLOYEE_DATA
```


What are the semi structured data types supported in Snowflake ?

VARIANT VARIANT data type is a superset of OBJECT and ARRAY.

It can store both OBJECT and ARRAY data either individually or combined together.

OBJECT + ARRAY= JSON

```
CREATE TABLE EMPLOYEE_DATA  
(  
  EMPLOYEE_INFO VARIANT  
);
```

```
{  
  "employee_id": "105",  
  "employee_name": "David Austin",  
  "position": "IT Programmer",  
  "phone_numbers": ["555-987-1234", "847-257-8594"],  
  "address": {  
    "street": "2014 JABBERWOCKY RD",  
    "city": "SOUTHLAKE",  
    "state": "TX",  
    "zip_code": "26192"  
  },  
  "skills": [  
    {  
      "skill_name": "Project Management",  
      "proficiency_level": "Expert"  
    },  
    {  
      "skill_name": "Agile Methodologies",  
      "proficiency_level": "Advanced"  
    },  
    {  
      "skill_name": "Python",  
      "proficiency_level": "Expert"  
    },  
    {  
      "skill_name": "DataBricks",  
      "proficiency_level": "Expert"  
    }  
  ]  
}
```

Summary

- Semi-Structured data is a form of data that does not conform to a rigid, predefined data model like a table with fixed structure but rather contains properties such as Key value pairs(JSON) or Tags (XML) to store or represent the data.
- Semi-structured data types supported by Snowflake are JSON, Parquet, Avro, ORC and XML.
- XML and JSON can be used for storing and transmitting data between applications.
- JSON uses Key Value pairs to store data and the key can be used to identify and select data from the JSON file.
- There are 3 semi-structured data types that are supported by Snowflake
 - OBJECT
 - ARRAY
 - VARIANT
- An Object consists of one or more key value pairs where each key is a unique string and value can be stored for this unique string or Key.
- An ARRAY is an ordered collection of elements and stores a list of values. The elements in the array have a specific sequence and you can access each element by its position (index) in the array.
- VARIANT data type is a superset of OBJECT and ARRAY. It can store both OBJECT and ARRAY data
- Object data can be identified by curly brackets.
- Array data can be identified by square brackets.
- OBJECT + ARRAY= JSON and is stored in VARIANT data type.

Summary

- JSON file format is needed to parse OBJECT, ARRAY or JSON data when loading.
- When creating a stage to load JSON data a JSON file format can be used.
- The elements of the JSON can be accessed in a SELECT statement by using column name and colon(:)
- The elements of a nested JSON can be accessed in a SELECT statement by using a dot(.) after the parent JSON column name.
- Data from the JSON SELECT will contain double quotes and these can be removed by casting to String
- ARRAYS are accessed by using Index , ARRAYS start at index 0 and do not have a limit to the number of elements it can have.
- Use FLATTEN to convert ARRAY elements into rows.
- An ARRAY with 100 elements will have 100 rows after being flattened with FLATTEN.
- JSON files may have a single JSON record in it or multiple JSON records, if the file has multiple JSON records then each JSON record will be part of an ARRAY.

Thank you



Learn2
CloudData

Powered by

www.learn2clouddata.com

Find Us

www.learn2clouddata.com

+01 234 567 89